

Vector-Space Optimization for Contraction Theory-Based Control Design: An Energy-Based Effective Space Approach [★]

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Abstract: In this paper, we propose a computationally efficient and numerically robust contraction metric calculation method for contraction theory-based control design. Conventional Linear Matrix Inequality (LMI) optimization frameworks for contraction metric calculation suffer from high computational complexity for real-time implementation and scale-invariance issues of the objective function which minimizes the condition number of the contraction metric. To address these issues, we project the contraction metrics onto the trajectory error vector space, leading to a simplified vector-space optimization framework within a lower-dimensional effective space. Furthermore, energy-based constraints are incorporated to obtain a unique solution and to maintain sufficient control effort. In addition, convex input saturation constraints are integrated to handle practical limitations of the actuators. The effectiveness and the feasibility of the proposed method are validated through numerical simulations using the Lorenz system.

Keywords: Contraction theory, input saturation, energy-based control

1. INTRODUCTION

1.1 Motivation

Besides the traditional Lyapunov stability theory, contraction theory offers an interesting perspective on the stability of nonlinear systems. It allows us to analyze the system stability by examining the convergence of trajectories, not the convergence to a specific equilibrium point Lohmiller and Slotine (1998). Since the contraction property is defined in an differential sense, the Linear Time-Varying (LTV) system analysis techniques can be applied to the contraction theory-based design.

Among recent developments in contraction theory-based control, Tsukamoto et al. proposed ConVex optimization-based Steady-state Tracking Error Minimization (CV-STEM) to systematically calculate the contraction metrics in Linear Matrix Inequalities (LMIs) optimization problem Tsukamoto and Chung (2021), using the differential nature of contraction theory.

However, since the LMI framework optimizes the contraction metrics in the matrix space, the computational complexity is $\mathcal{O}(n^6)$ [...], which can be a critical issue for high-dimensional systems. Thereby, for the real-time implementation, it is necessary to approximate the contraction metrics using universal function approximators such as neural networks [...].

Moreover, there is no consideration of input saturation in the conventional contraction theory-based control design which limits the practical applicability of the method.

To the best of our knowledge, there has been no previous work that projects the contraction constraints onto a vector space to reduce the computational complexity of the contraction metric calculation. Moreover, input saturation has not been considered in the conventional contraction theory-based control design. Hence, the main objective of this paper is to simplify the contraction metric calculation from matrix-space optimization to vector-space optimization, while also incorporating the input saturation constraint.

1.2 Contributions

The main contributions of this paper are

- The contraction metrics can be calculated in a lower-dimensional effective space by projecting the contraction constraints onto the trajectory error vector space, which significantly reduces the computational complexity of the LMI framework for contraction metric calculation;
- Scale-invariance issues of the objective function in the conventional LMI framework are resolved by introducing energy-based constraints that prevent the contraction metrics from collapsing or exploding, which ensures minimum required control effort;
- Saturation constraint is incorporated into the convex optimization framework, which handles practical limitations of the actuators; and

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- Robust feasibility of the proposed method is enhanced by ... slack variables

2. PRELIMINARIES

Before presenting the proposed method, we first review the contraction theory and related theorems. The detailed explanation of contraction theory can be found in Lohmiller and Slotine (1998); Tsukamoto et al. (2021b).

2.1 Contraction Property and Contraction Metric

Consider that we have the following deterministic nonlinear system:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t), \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^n$ denotes the state variables and $\mathbf{f} := \mathbf{f}(\mathbf{x}, t)$ denotes the nonlinear system function. Additionally, consider differential dynamics of the system (1) (Kirk, 2004, Chap 4), i.e., $\frac{d}{dt}\delta\mathbf{x} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\delta\mathbf{x}$.

According to [...], the system (1) is said to be *contracting*, i.e., all trajectories converge incrementally exponentially to a unique trajectory, if the following condition holds:

$$\frac{d}{dt}\mathbf{M} + 2\text{sym}\left(\mathbf{M}\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right) \preceq -2\alpha\mathbf{M}, \quad (2)$$

where $\mathbf{M}$ denotes *contraction metric* and $\alpha \in \mathbb{R}_{>0}$ denotes the contraction rate, and $\text{sym}(\mathbf{A}) := \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ Tsukamoto et al. (2021a). The contraction metric $\mathbf{M}$ is symmetric positive definite matrix and satisfies $\bar{m}\mathbf{I}_n \preceq \mathbf{M} \preceq \underline{m}\mathbf{I}_n$ with $\bar{m}, \underline{m} \in \mathbb{R}_{>0}$.

It is notable that the initial conditions are exponentially forgotten in the contracting system Lohmiller and Slotine (1998).

2.2 Perturbed Contracting System

If the system (1) is subjected to bounded disturbances, the system can be expressed as follows:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{d}(t), \quad (3)$$

where $\mathbf{d}(t) \in \mathbb{R}^n$ denotes the external disturbances with $\|\mathbf{d}(t)\| \leq \bar{d}$ where $\bar{d} \in \mathbb{R}_{>0}$.

The following theorem states that all solutions of the perturbed contracting system (3) exponentially converge to a bounded ball of the solution of the nominal contracting system (1).

Theorem 1. If the system (1) is contracting with the contraction metric $\mathbf{M}$ which satisfies the condition (2), the path integral $V_l(\mathbf{q}, \delta\mathbf{q}, t) = \int_{\xi_1}^{\xi_2} \|\Theta(\mathbf{q}, t)\delta\mathbf{q}\|$, where $\Theta(\mathbf{q}, t) := \mathbf{M}^{1/2}$, and ξ_1 and ξ_2 are the solutions of (1) and (3), respectively, and $\mathbf{q}$ is the virtual state variable, exponentially converges to a bounded ball. Especially, the following inequality of a bounded error ball of the solutions of (1) and (3) holds:

$$\|\xi_1(t) - \xi_2(t)\| \leq \frac{V_l(0)}{\sqrt{\bar{m}}} \exp(-\alpha t) + \frac{\bar{d}}{\alpha} \sqrt{\chi} (1 - \exp(-\alpha t)), \quad (4)$$

where $\chi := \bar{m}/\underline{m}$ denotes the condition number of $\mathbf{M}$, yielding the steady state error bound of $\frac{\bar{d}}{\alpha}\chi$.

Proof. See the proof of Theorem 2.4 in Tsukamoto et al. (2021b).

3. PROBLEM FORMULATION

Let us consider the following nonlinear system as follows:

$$\frac{d}{dt}\mathbf{x} = \mathbf{f}(\mathbf{x}, t) + \mathbf{B}(\mathbf{x}, t) \text{sat}(\mathbf{u}) + \mathbf{d}(t), \quad (5)$$

where $\mathbf{x} \in \mathbb{R}^n$ denotes the state variables, $\mathbf{u} \in \mathbb{R}^m$ denotes the control inputs, and $\mathbf{d} := \mathbf{d}(t) \in \mathbb{R}^n$ denotes the bounded external disturbances, i.e., $\|\mathbf{d}\| \leq \bar{d}$, and $\mathbf{f} := \mathbf{f}(\mathbf{x}, t)$ and $\mathbf{B} := \mathbf{B}(\mathbf{x}, t)$ denote system functions. The input saturation function is denoted by $\text{sat}(\cdot)$ and can be defined as any convex function which is physically feasible.

We assume that some desired control input $\mathbf{u}_d \in \mathbb{R}^m$ is pre-designed to achieve the desired trajectory $\mathbf{x}_d \in \mathbb{R}^n$ under ideal conditions, i.e., $\mathbf{d} = 0$ and $\|\mathbf{u}_d\| \leq \bar{u}$ where $\bar{u} \in \mathbb{R}_{>0}$ denotes the maximum allowable input magnitude by simulating the following desired system:

$$\frac{d}{dt}\mathbf{x}_d = \mathbf{f}_d + \mathbf{B}_d\mathbf{u}_d, \quad (6)$$

where $\mathbf{f}_d := \mathbf{f}(\mathbf{x}_d, t)$ and $\mathbf{B}_d := \mathbf{B}(\mathbf{x}_d, t)$.

The error dynamics between the system (5) and the desired system (6) can be expressed as follows:

$$\frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) = \mathbf{f} - \mathbf{f}_d + \mathbf{B} \text{sat}(\mathbf{u}) - \mathbf{B}_d\mathbf{u}_d + \mathbf{d}. \quad (7)$$

Using the state-dependent coefficient (SDC) formulation Dani et al. (2015), i.e., $\mathbb{A} := \int_0^1 \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right] (c\mathbf{x} + (1-c)\mathbf{x}_d) dc$, the error dynamics (7) can be linearized as follows:

$$\begin{aligned} \frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) &= \underbrace{\mathbf{f} - \mathbf{f}_d + (\mathbf{B} - \mathbf{B}_d)\mathbf{u}_d}_{=\mathbb{A}(\mathbf{x}, \mathbf{x}_d)(\mathbf{x} - \mathbf{x}_d)} \\ &\quad + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d} \\ &= \mathbb{A}(\mathbf{x} - \mathbf{x}_d) + \mathbf{B}(\mathbf{u} - \mathbf{u}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d} \end{aligned} \quad (8)$$

To simplify the control design, we use the following state feedback control law:

$$\mathbf{u} = \mathbf{u}_d - \mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M}(\mathbf{x} - \mathbf{x}_d), \quad (9)$$

where $\mathbf{R} \in \mathbb{R}^{m \times m}$ denotes a positive definite weighting matrix for the control input. Substituting the control law (9) into the error dynamics (8), we obtain:

$$\frac{d}{dt}(\mathbf{x} - \mathbf{x}_d) = \left(\mathbb{A} - \mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \right) (\mathbf{x} - \mathbf{x}_d) + \mathbf{B}\phi(\mathbf{u}) + \mathbf{d}, \quad (10)$$

where $\phi(\mathbf{u}) := \text{sat}(\mathbf{u}) - \mathbf{u}$.

Finally, the control objective is to obtain the contraction metric $\mathbf{M}$ in (9) which guarantees the system (5) to be contracting in the presence of input saturation.

4. CONVENTIONAL LMI-APPROACH

In [...], Tsukamoto et al. proposed ConVex optimization-based Steady-state Tracking Error Minimization (CV-STEM) to obtain the contraction metric $\mathbf{M}$ by solving the following LMI optimization problem. Ignoring the input saturation effect, the contraction constraint of the error dynamics (10) can be expressed as follows:

$$\frac{d}{dt}\mathbf{M} + 2\text{sym}(\mathbf{M}\mathbb{A}) - 2\mathbf{M}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\top \mathbf{M} \preceq -2\alpha\mathbf{M}. \quad (11)$$

Using the variable transformation $\mathbf{M} =: \mathbf{W}^{-1}$, we have the following LMI optimization problem:

$$\begin{aligned} & \min_{\bar{\mathbf{W}}, \chi, \nu} \chi + \lambda \nu \\ \text{s.t. } & \begin{cases} -\frac{d}{dt} \bar{\mathbf{W}} + 2 \text{sym}(\mathbf{A} \bar{\mathbf{W}}) - 2\nu \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^\top \preceq -2\alpha \bar{\mathbf{W}}, \\ \mathbf{I}_n \preceq \bar{\mathbf{W}} \preceq \chi \mathbf{I}_n, \end{cases} \end{aligned} \quad (12)$$

where $\bar{\mathbf{W}} := \nu \mathbf{W}$ and $\nu \in \mathbb{R}_{>0}$ is a penalty term of 2-norm of $\mathbf{M}$. Using $\lambda \in \mathbb{R}_{>0}$, the magnitude of the contraction metric $\mathbf{M}$ can be adjusted.

As well known, the computational complexity of solving the above LMI optimization problem is $\mathcal{O}(n^6)$ [...], which can be a critical issue for high-dimensional systems. In this regard, the conventional method is not suitable for real-time implementation.

Moreover, since CV-STEM penalizes the magnitude of the control input indirectly by adjusting $\mathbf{M}$, the input saturation constraint is not explicitly considered in the optimization problem. Thus, Theorem 1 cannot be directly applied to the system (5) with input saturation, since $\phi(\mathbf{u})$ is not considered as bounded disturbances.

In conclusion, though CV-STEM provides a systematic framework for contraction metric calculation in LMI framework, there still remain several issues to be resolved for practical real-time implementation.

Remark 2. There exist a severe problem formulation issue in the conventional LMI framework. In the formulation, $\frac{d}{dt} \bar{\mathbf{W}} = \frac{d}{dt} (\nu \mathbf{W})$ is considered as same as $\nu \frac{d}{dt} (\mathbf{W})$, which is not mathematically valid since ν is also a decision variable. Thus, the conventional LMI framework's solutions are typically decides $\bar{\mathbf{W}}$ and χ to $\mathbf{I}_n$ and 1, respectively, and modifies ν only.

5. MAIN RESULTS

5.1 Vector-Space Optimization Framework

Since the contraction metric $\mathbf{M}$ is only applied to the trajectory error vector $\mathbf{e} := \mathbf{x} - \mathbf{x}_d$ in the control law (9), we can project the contraction constraints onto the trajectory error vector space. By pre- and post-multiplying $\mathbf{e}^\top$ and $\mathbf{e}$ to the contraction constraint (11), respectively, we have:

$$\mathbf{y}^\top \left(-\mathbf{B} \mathbf{R}^{-1} \mathbf{B}^\top \right) \mathbf{y} + \mathbf{e}^\top \left(\frac{1}{T_s} + 2\alpha + 2\mathbf{A}^\top \right) \mathbf{y} - \frac{\mathbf{e}^\top \mathbf{M}_{\text{pre}} \mathbf{e}}{T_s} \leq 0, \quad (13)$$

where $\mathbf{y} := \mathbf{M} \mathbf{e}$ denotes a projected contraction metric on error vector and a new optimization variable. Note that $\frac{d}{dt} \mathbf{M} \approx \frac{1}{T_s} (\mathbf{M} - \mathbf{M}_{\text{pre}})$ where T_s denotes the sampling time and $\mathbf{M}_{\text{pre}}$ denotes the contraction metric obtained at the previous time step, for discrete-time implementation.

The convex input saturation function $\text{sat}(\cdot)$ preserves the convexity on $\mathbf{y}$, since $\mathbf{u}$ is linear with respect to $\mathbf{y}$. Thus, the input saturation constraint can be expressed as a convex constraint function of $\mathbf{y}$ as follows:

$$c(\mathbf{y}) = \text{sat}(\mathbf{u}) \leq 0. \quad (14)$$

According to Theorem 1, the steady-state error bound is proportional to the condition number of the contraction metric $\mathbf{M}$. We can indirectly minimize the condition number by minimizing the angle between $\mathbf{y}$ and $\mathbf{e}$, since the optimal condition number of $\mathbf{M}$ is achieved when $\mathbf{y}$ is parallel to $\mathbf{e}$, resulting $\mathbf{M} = \gamma \mathbf{I}_n$ for some $\gamma \in \mathbb{R}_{>0}$.

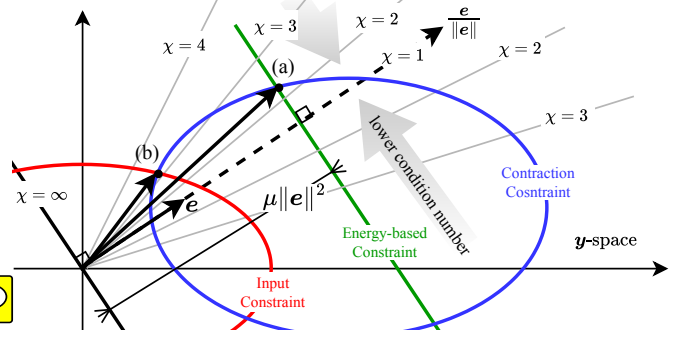


Fig. 1. Illustration of the optimization problem in two-dimensional space, see Section 5.2 for details.

Algorithm 1 Proposed Method

- 1: **Initialize:** $\mathbf{y}_0 = \mathbf{1}_{n \times 1}$, $s_0 = 0$, $\mathbf{M}_0 = \mu \mathbf{I}_n$
- 2: **for** $k = 1$ to ∞ **do**
- 3: Get current $\mathbf{x}_k$, $\mathbf{x}_{d,k}$, $\mathbf{u}_{d,k}$ and $\mathbf{e}_k$
- 4: Assign $\mathbf{y}_k \leftarrow \mathbf{y}_{k-1}$, $s_k \leftarrow s_{k-1}$ for warm-start
- 5: Solve (16) to get optimal $\mathbf{y}_k$ and s_k
- 6: Calculate $\mathbf{M}_k$ using $\mathbf{y}_k$ and $\mathbf{e}_k$ using (17)
- 7: Apply the control law (9) using $\mathbf{M}_k$
- 8: **end for**

However, there is infinite number of solutions $\mathbf{y}$ which is parallel to $\mathbf{e}$, i.e., since the condition number is scale-invariant. To this end, energy-based constraint is introduced to obtain a unique solution, maintaining sufficient control effort. Specifically, the following energy-based constraint is considered:

$$(\mathbf{e}^\top \mathbf{y} - \mu \mathbf{e}^\top \mathbf{e})^2 \leq (\epsilon + s) \|\mathbf{e}\|^2, \quad (15)$$

where $\mu \in \mathbb{R}_{>0}$ denotes the desired energy level, and $\epsilon \in \mathbb{R}_{>0}$ and $s \in \mathbb{R}_{\geq 0}$ denote allowable small energy variation and a slack variable to enhance the robust feasibility of the optimization problem.

Thus, the following vector-valued optimization problem can be formulated to obtain the projected contraction metric $\mathbf{y}$:

$$\begin{aligned} & \min_{\mathbf{y}, s} \|\mathbf{y}\| \|\mathbf{e}\| - \mathbf{y}^\top \mathbf{e} + \lambda s \\ & \text{s.t. } (13), (14), (15), \end{aligned} \quad (16)$$

After the optimization, the contraction metric $\mathbf{M}$ can be retrieved maintaining the condition number and the relationship $\mathbf{y} = \mathbf{M} \mathbf{e}$ as follows:

$$\mathbf{M} = \frac{1}{\sqrt{\chi}} \text{diag} \left(\left(\sqrt{\chi}, 1, \dots, 1, \frac{1}{\sqrt{\chi}} \right) \right) \cdot \frac{1}{\|\mathbf{e}\| \|\mathbf{y}\|}, \quad (17)$$

where $\chi = \frac{1 + \sin(\theta)}{1 - \sin(\theta)}$ denotes the minimum condition number of $\mathbf{M}$ when the angle θ between $\mathbf{y}$ and $\mathbf{e}$ is given.

The overall procedure of the proposed method is summarized in Algorithm 1 in which the optimization problem (16) is solved at each time step.

5.2 Illustrative Example

For better understanding, the optimization problem (16) is illustrated in two-dimensional space in Fig. 1. The contraction constraint (13) and the input saturation constraint (14) restrict the feasible space of $\mathbf{y}$ in the outside of blue ellipse and the inside of red ellipse, respectively. The green

Table 1. Controllers for comparative study.

Description	
(C ₁) [—]	Proposed controller.
(C ₂) [—]	CV-STEM (Tsukamoto and Chung (2021)).
(C ₃) [—]	Desired controller, i.e., $\mathbf{u} = \mathbf{u}_d$.

line represents the energy-based constraint (15) assuming that the allowable energy variation ϵ is approximately zero.

Neglecting the input saturation constraint, the optimal solution is obtained at the point (a) accordingly the objective function. Considering the input saturation constraint, the optimal solution is obtained at the point (b) by increasing the slack variable s . This also ensures that there exists a **unique** optimal solution in the optimization problem (16).

6. NUMERICAL VALIDATION

6.1 Numerical Validation Set Up

We employed the Lorenz system [MS: REF] to validate the proposed method which is represented as follows:

$$\underbrace{\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{=: \dot{\mathbf{x}}} = \underbrace{\begin{pmatrix} \sigma(y-x) \\ x(\rho-z)-y \\ xy-\beta z \end{pmatrix}}_{=: \mathbf{f}(\mathbf{x})} + \text{sat} \left(\underbrace{\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}}_{=: \mathbf{u}} \right) + \mathbf{d}(t), \quad (18)$$

where $\sigma = 10$, $\rho = 28$, and $\beta = \frac{8}{3}$ are system parameters. The input saturation function $\text{sat}(\cdot)$ was defined as follows:

$$\text{sat}(\mathbf{u}) := \begin{cases} \mathbf{u}, & \text{if } \|\mathbf{u}\| \leq \bar{u}, \\ \bar{u} \frac{\mathbf{u}}{\|\mathbf{u}\|}, & \text{otherwise,} \end{cases} \quad (19)$$

where $\bar{u} = 200$ denotes the maximum input magnitude. The external disturbances $\mathbf{d}(t)$ were modeled as follows:

$$\mathbf{d}(t) := 1000(H(t-0.1) - H(t-0.11)) \cdot \mathbf{1}_{3 \times 1}, \quad (20)$$

where $H(\cdot)$ denotes the Heaviside step function.

For the desired trajectory, we simulated the desired system (6) with the following desired feedback-linearization control input $\mathbf{u}_d := -\mathbf{f}_d - 2(\mathbf{x}_d - \mathbf{r})$, where $\mathbf{r} := (-8.485, 8.485, 27)^\top$ denotes the reference point.

The proposed method (C₁) was compared with CV-STEM (C₂) Tsukamoto and Chung (2021) and the desired controller, i.e., $\mathbf{u} = \mathbf{u}_d$, (C₃) as summarized in Table 1. The parameters shared by (C₁) and (C₂) were $\mathbf{R} = \mathbf{I}_3$ and $\alpha = 10$. In addition, parameters were selected as $\mu = 30$ and $\lambda = 100$ for (C₁) and $\lambda = 0.01$ for (C₂). To solve the optimization problems, `fmincon` function in MATLAB R2024b and `SeMeDi` toolbox were used for (C₁) and (C₂), respectively.

The simulation was conducted for $T = 0.5$ seconds with the sampling time of $T_s = 0.005$ seconds for the controllers and 10 times finer time step for the system simulation. The initial conditions were set to $\mathbf{x}(0) = (10, 5, 13)^\top$ and $\mathbf{x}_d(0) = (11, 4, 15)^\top$.

6.2 Numerical Validation Results

The numerical validation results are illustrated in Figs. 2 and 3. When only the desired controller was applied, i.e., (C₃), the trajectories of (C₃) and $\mathbf{x}_d$ does not coincide due

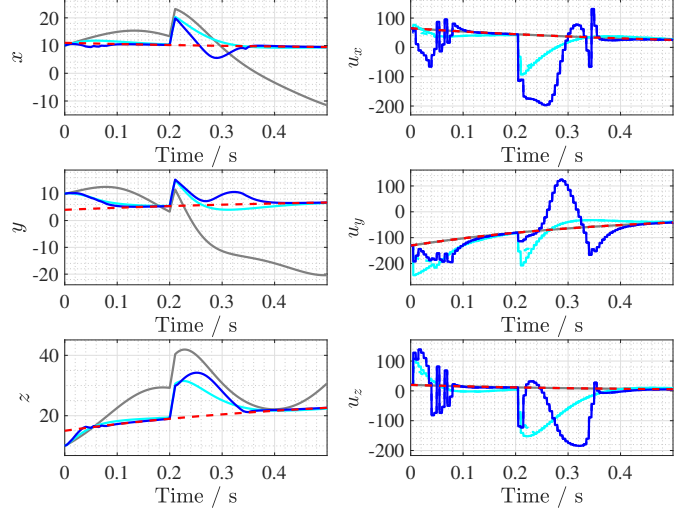


Fig. 2. Numerical validation results of (C₁) [—], (C₂) [—], (C₃) [—], and desired trajectory $\mathbf{x}_d$ [—]

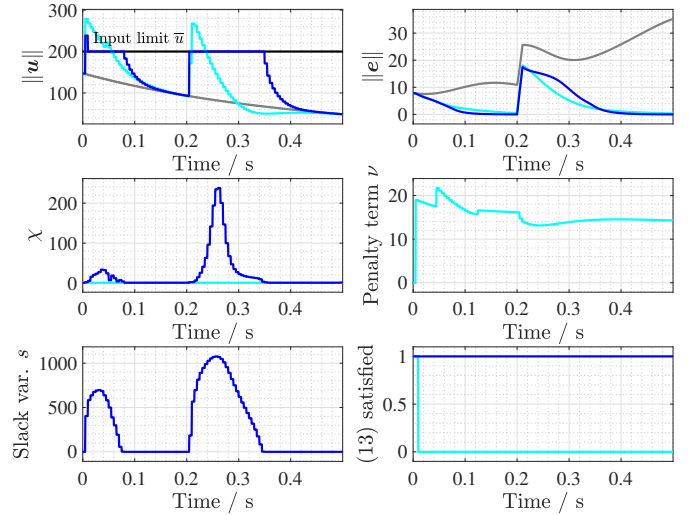


Fig. 3. Control input norms, condition numbers, penalty term and slack variable.

Table 2. Tracking performances in L_2 norm.

	(C ₁) [—]	(C ₂) [—]	(C ₃) [—]
$\int_0^T \ \mathbf{e}\ dt$	0.4937	0.7929	4.0534
	(-87.82%)	(-80.44%)	(-)

to the effect of initial condition mismatch. Significantly, after the disturbance occurrence at $t = 0.2$ seconds, the trajectory deviates further from the desired trajectory as shown in Fig. 2.

By applying the contraction-based controllers, both (C₁) and (C₂) successfully drive the trajectory to closely follow the desired trajectory $\mathbf{x}_d$ even in the presence of disturbances, minimizing L_2 norm of the tracking error about 87.82% and 80.44% compared to (C₃) as summarized in Table 2. However, in terms of the input saturation, (C₂) reaches the saturation limit by the initial condition mismatch and the disturbance occurs, while (C₁) effectively avoids input saturation as shown in Fig. 3. Moreover, (C₁) effectively tracked the desired trajectory faster than

(C_2) while satisfying the input saturation constraint. This demonstrates the effectiveness of the proposed method in handling input saturation constraints while achieving satisfactory tracking performance.

In addition, the expected condition number χ behaviors of the contraction metrics in Section 5 through Fig. 1 of (C_1) can be observed in Fig. 3. As the constraint constraint (11) and the input saturation constraint (14) become active, the angle between $\mathbf{y}$ and $\mathbf{e}$ increases, resulting in the increase of the condition number. Specifically, when the input saturation constraint became active, the condition number χ increased from 1 and decreased again to 1 after the saturation effect disappeared.

In contrast, (C_2) shows a smallest condition number, $\chi = 1$, due to the problem formulation issue discussed in Remark 2. Instead of adjusting the condition number, (C_2) increases the penalty term ν to increase the control effort as shown in Fig. 3, while making $\bar{\mathbf{W}}$ close to identity matrix, i.e., $\mathbf{M} = \nu \mathbf{I}_n$. This shows the issues of the conventional optimization formulation discussed in Remark 2.

7. CONCLUSION

In this study, we proposed a vector-space optimization framework for contraction metric calculation in contraction theory-based control design. By projecting the contraction constraints onto the trajectory error vector space, the contraction metrics can be calculated in a lower-dimensional effective space, significantly reducing the computational complexity of the conventional LMI framework.

The objective function to minimize the steady-state error bound was reformulated to minimize the angle between the projected contraction metric and the trajectory error vector, which minimizes the condition number of the contraction metric. To resolve scale-invariance issues of the objective function, i.e., infinite number of solutions yielding the same condition number, energy-based constraints were introduced maintaining sufficient control effort. Additionally, in the vector-space, convex input saturation constraints were easily incorporated into the proposed framework.

The effectiveness and the feasibility of the proposed method were validated through numerical simulations using the Lorenz system.

As a future work, we will extend the proposed method to handle state constraints and validate through real-world experiments.

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